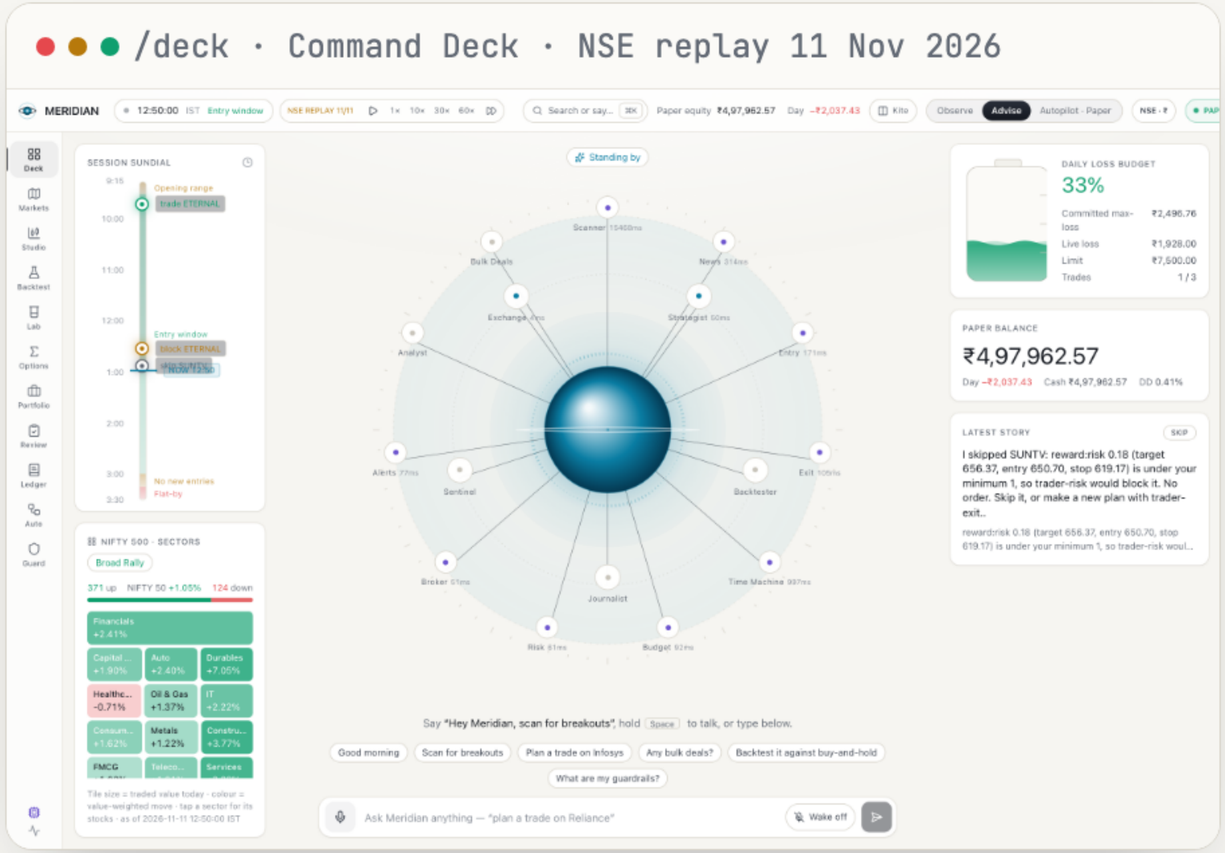
PS-02 · AGENTIC TRADING BOT · PAPER ONLY

MERIDIAN

Speak the market.
Trust every trade.

A voice-first paper-trading desk for India's NSE. One Conductor listens and drives sixteen agents; five defence layers stand between every idea and every order, and every decision gets a plain-English story.

16	5	497	₹0
agents	defence layers	NSE stocks scanned	real money



"Hey Meridian, scan for breakouts"

"Zomato mein trade plan karo"

"Stop everything"

Traders know the rules. Nothing makes them follow them.

WHAT SEBI FOUND

7 in 10

individual intraday equity traders lost money in FY23 (SEBI study, July 2024)

93%

of individual F&O traders made losses over FY22–FY24 (SEBI study, Sept 2024)

Root causes: entries on noise, rules kept in the head, size set by emotion, revenge trades after a loss, and no record of *why*.

Moment	Before: a typical retail day	After: with Meridian
Finding a trade	Tips, charts, FOMO	Scanner ranks 497 NSE stocks on one rule
Sizing	"Feels right"	Budget sizes to a ₹2,500 max loss
Stop-loss	Often skipped or moved	Required before entry, OCO bracket
After a loss	Revenge re-entry	Tilt guard: 15-minute cool-down
Limits	Raised mid-day	Voice can only tighten them
Afterwards	No idea why	A Story and a post-mortem per trade

You talk
English or Hinglish, voice or ⌘K

Agents work
Scan, plan, test, size, gate

Rules hold
Code, not willpower

Sources: SEBI studies on intraday (Jul 2024) and F&O (Sep 2024) trader outcomes

One intelligence, sixteen tools, policy outside the model

Experience — Landing · Deck · Markets · Studio · Lab · Options · Review · Guardrails

Voice gateway — Wake word · VAD · Whisper STT · Hinglish · TTS

Conductor — intent grammar → planner → narrator

- Policy engine
- Tilt guard
- Automations
- Compliance

11 skill agents

Scanner · News · Entry · Exit · Time Machine · Budget · Risk · Broker · Alerts · Analyst · Bulk Deals

5 native services

Strategist · Backtester · Journalist · Sentinel · Paper Exchange

State and data — SQLite events · hash-chained audit · real NSE replay · Yahoo .NS

TECH STACK

Layer	Built with
Frontend	Next.js 16, React 19, Tailwind 4
Motion, 3D	Motion 12, three.js
Charts	lightweight-charts, SVG
Backend	Python 3.12, FastAPI, WebSocket
Agents	day-trader-skill, India edition
Voice	faster-whisper, OmniVoice
LLM	Local Ollama (qwen2.5), optional
Storage	SQLite WAL, SHA-256 chain
Reports	ReportLab PDFs
Testing	pytest, Playwright

The LLM has no order-placing tool. Every order passes the Policy engine and a fingerprinted Risk PASS.

The guardrails held on a real NSE session

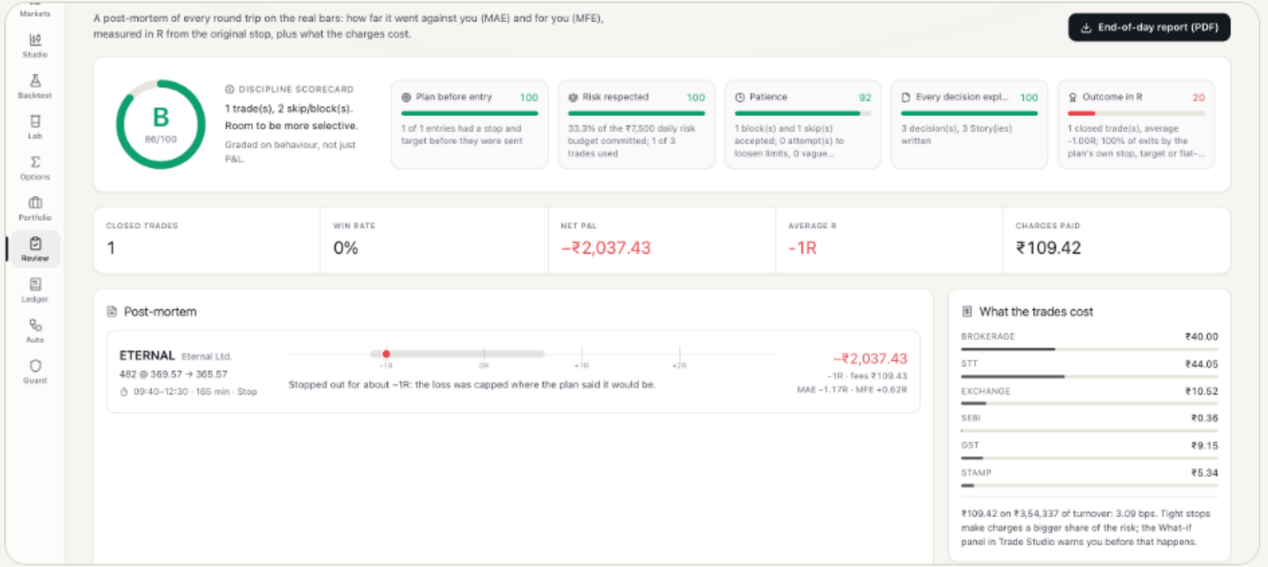
Risk on one ETERNAL trade, before and after

"Buy 600 Zomato", as asked — ₹3,108 · BLOCKED

482 shares, sized by Budget — ₹2,497 · PASS

Revenge re-entry after the stop — ₹0 · cool-down

Bar length = maximum loss at the stop on the recorded 11 Nov 2026 session. Per-trade cap ₹2,500; daily cap ₹7,500.



Learners

Practise with real data and no money at risk

Busy traders

Hands-free scans, automations, alerts

Brokers, colleges

A safe sandbox with a SEBI-style audit trail

117 unit tests pass

25/25 end-to-end checks

100% decisions explained

All figures from the prototype on the recorded NSE session; ₹ values include brokerage, STT, exchange, SEBI, GST and stamp duty.

Runs on a laptop today. Ready for a broker tomorrow.

TECHNICAL

Built and tested
13 screens, 50+ API routes, 117 tests, 25-check replay

Laptop CPU
Local STT, optional local LLM, no GPU needed

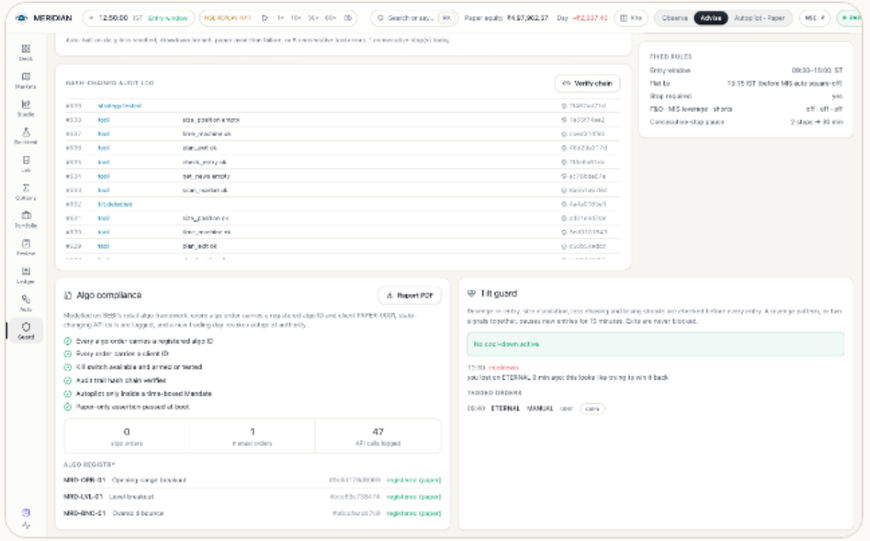
Real data
Recorded NSE session; live Yahoo .NS mode

ECONOMIC AND REGULATORY

Near-zero run cost
MIT and Apache-2.0 parts, free data, free local LLM

SEBI-aligned
Algo IDs, kill switch, API audit log, daily logout

Revenue paths
Free for learners; licensed to brokers and colleges



Risk	Mitigation
Unofficial data	Replay of a recorded day
LLM hallucination	Numbers validated vs tools
Real-money misuse	No live-broker code path

Roadmap: options chain feed · broker MCP read-only sync · Hindi speech recognition by default · mobile PWA

Where Meridian sits, and what it is built on

Capability	Meridian	No-code algo	Options tools	Journals
Voice, English + Hinglish	Yes	No	No	No
Rules enforced before entry	Yes	Partly	No	No
Tilt guard with cool-down	Yes	No	No	Flags only
Story for every decision	Yes	No	No	Manual
Honest backtest vs hold	Yes	Yes	Partly	No
Algo IDs + audit chain	Yes	Broker-side	No	No

Comparison from public product descriptions (no-code algo builders such as Streak, options analytics such as Sensibull, and journals such as Edgewonk). Category-level, not a product audit.

REFERENCES

SEBI · intraday trader study (Jul 2024); F&O study (Sep 2024)

SEBI · retail algo trading circular (4 Feb 2025)

day-trader-skill · 11 trading skills (MIT)

k2-fsa OmniVoice · multilingual TTS (Apache-2.0)

faster-whisper · local speech recognition

Data · NSE India archives, Yahoo Finance, Google News